

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Civil Engineering

Semester : IV

Subject Code & Name: BTCVC403 & Structural Mechanics - I

Max Marks: 60

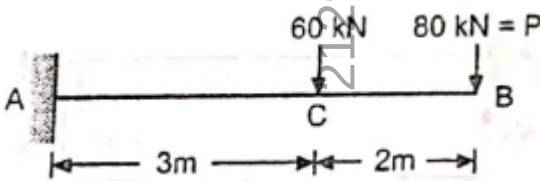
Date: 24/12/2024

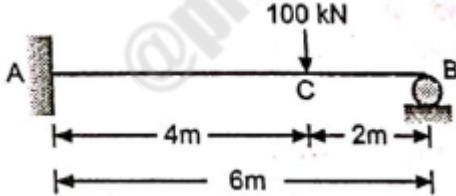
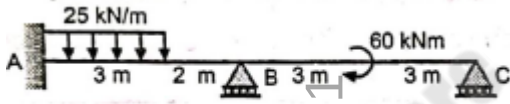
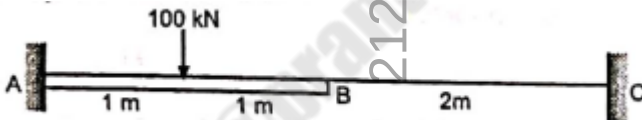
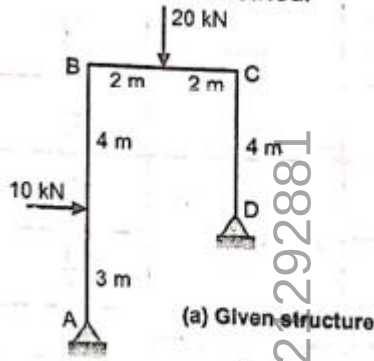
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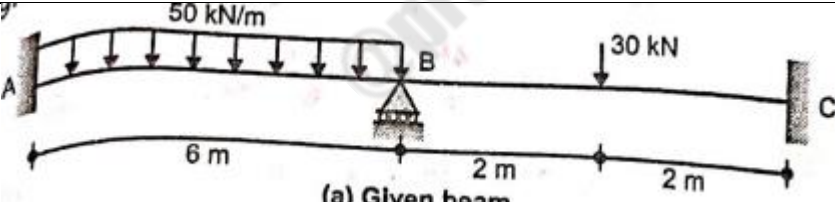
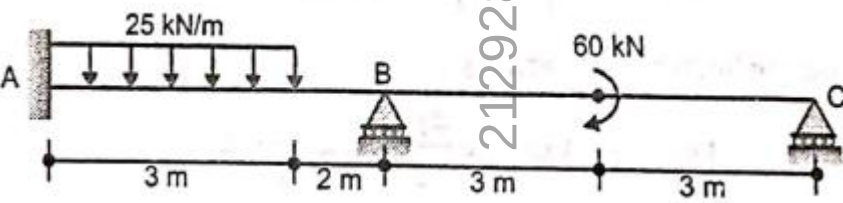
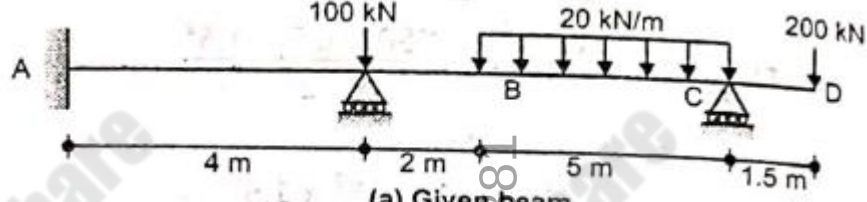
Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Mark s
Q. 1	Objective type questions. (Compulsory Question)		12
1	In cantilever beams, the slope is _____ at fixed end. a. Maximum b. Zero c. Minimum d. Uniform	CO2	1
2	Using Mohr's theorem, calculate the maximum slope of a cantilever beam if the bending moment area diagram is 90kNm ² . Take EI = 4000 kNm ² . a. 0.0225 radian b. 0.0367 radian c. 0.0455 radians d. 0.066 radians	CO2	1
3	In cantilever beam the deflection occurs at _____. a. Free end b. Point of loading c. Through out d. Fixed end	CO2	1
4	Compute the maximum deflection at free end of a cantilever beam subjected to udl for entire span of l metres. a. $wl^4/8EI$ b. $wl^4/4EI$ c. $wl^3/8EI$ d. $wl^2/6EI$	CO2	1
5	Calculate the maximum deflection of a cantilever beam with udl on entire span of 3m the intensity of you udl be 25 kN/m. Take EI as 4000 kNm ² . a. 0.052m b. 0.063m c. 0.076m d. 0.09m	CO2	1
6	Which of the following is statically determinate structure? a. Two hinged arch b. Fixed beam c. Double overhanging d. Continuous beam	CO1	1
7	While using three moment equation, a fixed end of a continuous beam is replaced by an additional span of a. Zero length b. Infinite length c. Zero moment of inertia d. None of the above	CO2	1

8	In moment distribution method, the sum of distribution factors of all the members meeting at any joint is always				CO2	1
	a. Zero	b. Less than 1	c. 1	d. Greater than 1		
9	The moment distribution method is best suited for				CO2	1
	a. Indeterminate pin jointed truss	b. Rigid frame	c. Space frames	d. Composite structure		
10	A beam AB of length 'L' is hinged at its ends and carries a transverse external loading such that the end 'B' is sunk by an amount 'δ'. The fundamental slope deflection equation is				CO2	1
	a. $M_{AB} = \frac{2EI}{L} (\theta_A + 2\theta_B - 3\delta/L) + M_{FAB}$	b. $M_{AB} = \frac{3EI}{L} (2\theta_A + \theta_B - 3\delta/L) - M_{FAB}$	c. $M_{AB} = \frac{3EI}{L} (2\theta_A + \theta_B + 3\delta/L) - M_{FAB}$	d. $M_{AB} = \frac{2EI}{L} (2\theta_A + \theta_B - 3\delta/L) + M_{FAB}$		
11	Castigliano's first theorem is applicable				CO2	1
	a. A For statically determinate structures only	b. When the system behaves elastically	c. Only when principle of superposition is valid	d. None of the above		
12	Clapeyron's theorem is also known as the theory of				CO2	1
	a. 3 - moments	b. 2 - moments	c. Single moment	d. None of above		
Q. 2	Solve the following.					12
A)	List out the different methods to find slope and deflection and explain them in detail?				CO2	6
B)	Derive an expression for slope and deflection for simply supported beam with a UDL throughout span using double integration method?				CO2	6
Q.3	Solve the following.					12
A)	Determine the maximum deflection for the beam as shown below. Assuming $E= 2 \times 10^5 \text{ N/mm}^2$, $I=1 \times 10^9 \text{ mm}^4$. Using Castiglianos First Theorem.				CO2	6
						
B)	Determine reaction at B for the propped cantilever AB as shown in fig below. Assuming constant flexural rigidity (EI)				CO2	6

			
Q. 4	Solve Any Two of the following.		12
A)	Determine the fixed end moments for eccentric point load case.	CO2	6
B)	A fixed beam of 2m span is carrying a point load of 50kN at its mid-point. Find the fixed end moment and deflection of the beam at its midpoint.	CO2	6
C)	A beam ABC is simply supported at A and C and is continuous over B. It is loaded with UDL intensity of load on AB is 60kN/m having span 4m and on BC 100kN/m having span 6m. Find moment at B and draw SFD and BMD.	CO2	6
Q.5	Solve Any Two of the following.		12
A)	Analyse the continuous beam ABC as shown in fig below by moment distribution method.	CO2	6
			
B)	Analyse the continuous beam ABC as shown in fig below by moment distribution method.	CO2	6
			
C)	Analyse the continuous beam ABC as shown in fig below by moment distribution method.	CO2	6
	 (a) Given structure		
Q. 6	Solve Any Two of the following.		12
A)	Analyse the continuous beam ABC as shown in fig below by slope deflection method.	CO2	6

	 (a) Given beam		
B)	Analyse the continuous beam ABC as shown in fig below by slope deflection method. 	CO2	6
C)	Analyse the continuous beam ABCD as shown in fig below by slope deflection method. The support B sinks by 15mm. $EI = 3800 \text{ kNm}^2$.  (a) Given beam *** End ***	CO2	6